Section 3.2). The current work does not contain such analysis and instead handles the goodness of the learned T_n by a single assumption. In particular, it would be of interest to theoretically justify the assumed "first order condition," i.e., the smallness of the $\mathcal{W}_2$-gradient ξ_{n+1} in (30), by analyzing the convergence of the optimization. One possibility is by showing the weak convergence of the learned p_{n+1} to the exact minimizer p^* of F_{n+1} and then utilizing the convergence of $\nabla_{\mathcal{W}_2} F_{n+1}(p)$ to $\nabla_{\mathcal{W}_2} F_{n+1}(p^*) = 0$ in a proper sense [7, Section 5.4]. Second, the current generation result only covers the case of G being the KL divergence. An extension to the cases when G is other types of divergence, such as f-divergence (see Remark 5.1), will broaden the scope of the result. Third, our theory uses the population quantities throughout. A finite-sample analysis, which can be based on our population analysis, will provide statistical convergence rates in addition to the current result.

Meanwhile, the JKO scheme computes a fully backward proximal GD. Given the existing convergence rates of the other Wasserstein GD [40, 63], one would expect that a variety of first-order Wasserstein GD schemes can be applied to progressive flow models and the theoretical guarantees can be derived similarly to the JKO scheme. We also note the connection between the JKO scheme and learning of the score function, at least in the limit of small step size [74, Section 3.2]. Given the growing literature on the analysis of score-based diffusion models, it can be worthwhile to investigate this connection further to develop new theories for the ODE flow models.

Finally, it would be interesting to use theory to guide practice and to develop new or improved methodologies of flow-based generative models. As discussed in Section 3.3, one may consider incorporating the inversion error as part of training loss to enforce the accuracy of the reverse process. The potential theoretical extension to f-divergences also suggests utilizing more general f-divergence as the per-step training objective in JKO flow networks, e.g., by adopting techniques in f-GAN [57]. It would be interesting to explore different choices of f and the relationships among the f-divergences [31]. In addition, our theory indicates that using larger step-size γ leads to a shorter sequence of Residual Blocks in the network architecture (as long as the optimization in each JKO step can be efficiently solved). It would be natural to consider an adaptive choice of γ in practice and also in extending the theory.

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